# Artificial Intelligence

Course Code: CS-202

Introduction to Artificial Intelligence Approaches

Instructor: Habiba Arshad

# OBE

## CLO Covered in this Lecture:

CLO1: **Discuss** the key concepts in the field of artificial intelligence

# Approaches to Building Intelligent Systems

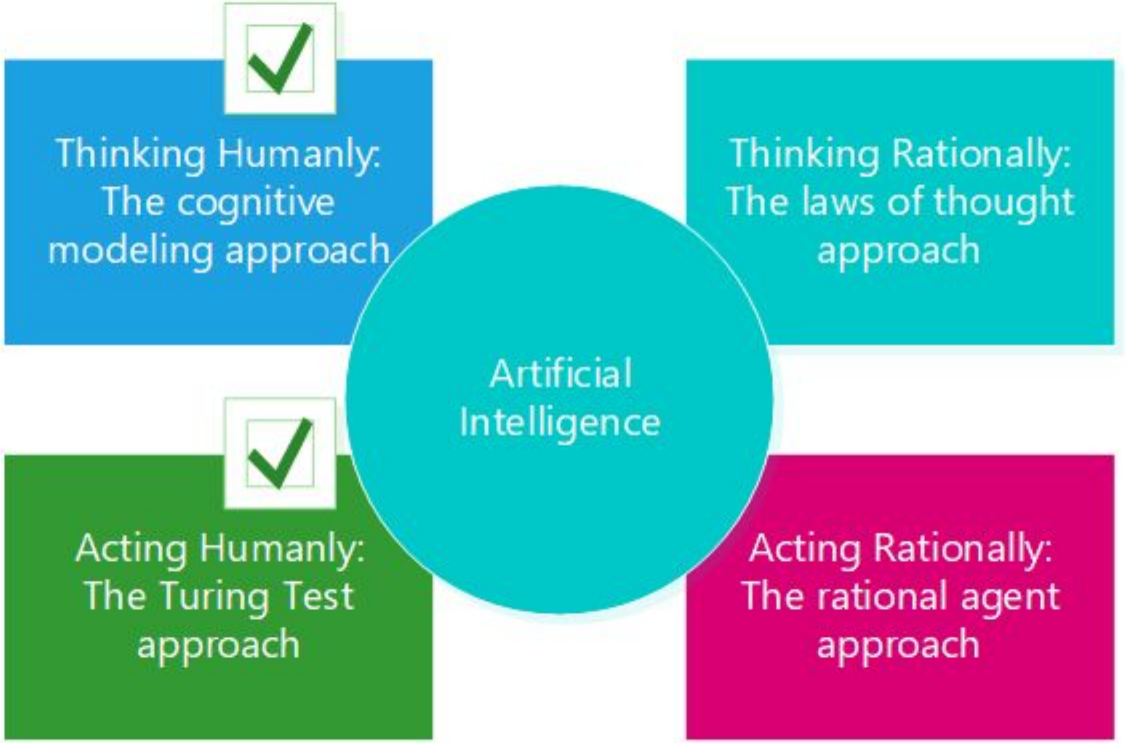
- Artificial Intelligence systems can be developed using different approaches depending on how they process information and make decisions. The two main approaches are:
- **Rule-Based AI (Knowledge-Driven Systems):**
  - Where a system operates under a defined set of rules and conditions
  - It follows “if–then” conditions to make decisions.
  - It does not learn from data
  - **Example:** Expert systems, chatbots with fixed responses, or a medical diagnosis system that follows a set of predefined conditions.

# Approaches to Building Intelligent Systems

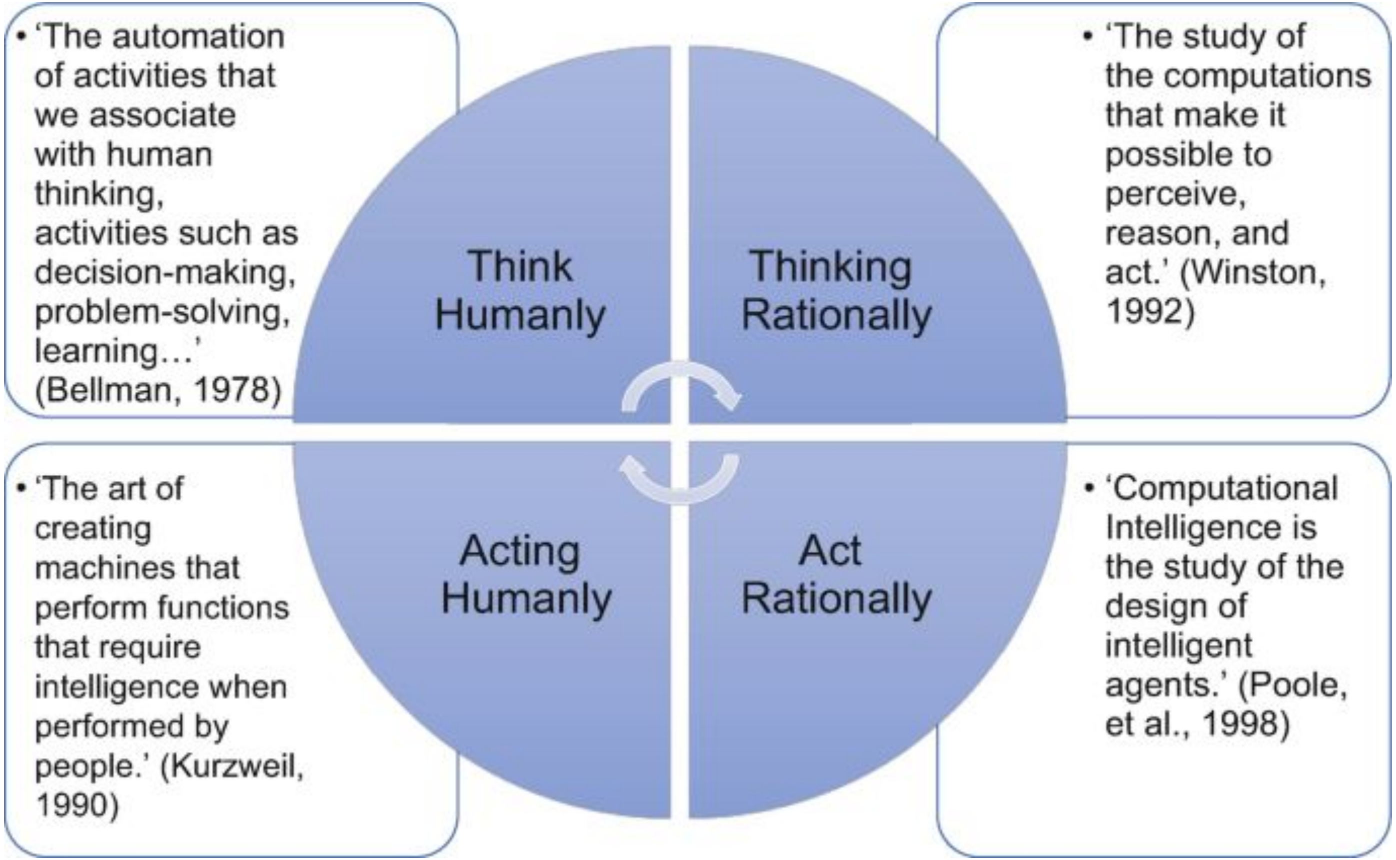
- **Learning-Based AI (Data-Driven Systems):**
  - where it improves through data and experience.
  - Instead of fixed rules, it learns patterns and relationships from data.
  - It can adapt and make predictions or decisions on new, unseen data.
  - **Example:** Image recognition systems, speech recognition, or recommendation engines like Netflix and YouTube.

# Four Main Approaches to Artificial Intelligence

How to make an AI model think and act like a human?



# Definitions for Approaches to Artificial Intelligence

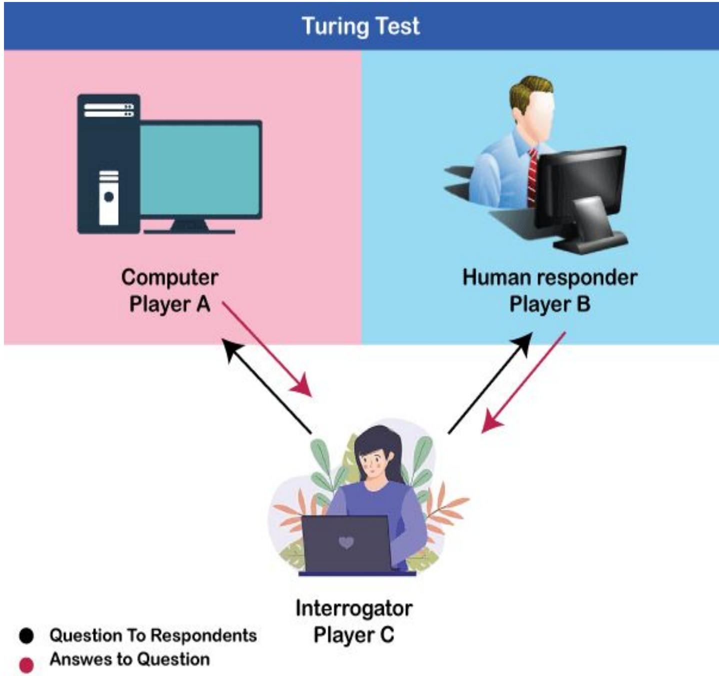


# **Approach 1:** Acting Humanly

- This approach aims to build systems that **behave like humans**, regardless of whether they think like humans internally. The goal is to make machines act intelligently in the way humans would in the same situation.
- **Example:**
- The **Turing Test** (by Alan Turing) measures a machine's ability to act humanly — if a human can't tell it's a machine, it passes.
- **Voice assistants** (like Siri or Alexa) that respond naturally to human speech.
- **Humanoid robots** that perform tasks such as greeting or assisting people.

# Turing Test

- Turing test: In 1950, Alan Turing introduced this method in his paper, "Computing Machinery and Intelligence," which considered the question, "Can machines think?“as a test to check whether or not a computer is capable of thinking like a human being.
  - Computer and the human both interrogated by the judge
  - Computer passes the test if judge can’t tell the difference



# How effective is this test?

**For an agent to pass the Turing Test, it must:**

- Have **command of language** (understand and respond naturally).
- Possess a **wide range of knowledge** to discuss various topics.
- **Demonstrate human traits** such as humor, emotions, and personality.
- Be able to **reason logically** and make appropriate decisions.
- Be able to **learn from experience** to improve responses over time.

# How effective is this test?

**Loebner Prize Competition**

The **Loebner Prize** is a **modern version of the Turing Test**, held annually to evaluate conversational AI programs (chatbots).

Programs are judged on how closely their responses resemble those of humans.

**Example:** *ALICE* (Artificial Linguistic Internet Computer Entity) won the Loebner Prize in **2000** and **2001** for its advanced conversational abilities.

# Approach 2: Thinking Humanly

- This approach focuses on making machines **think like humans** — by studying how people reason, learn, and make decisions. It uses concepts from psychology and cognitive science to model human thought processes.
- **Example:**
- Cognitive modeling — like simulating how the human brain solves puzzles.
- **Chatbots** that try to imitate human conversation or reasoning patterns.
- **Neural networks** designed to mimic how neurons in the brain work.

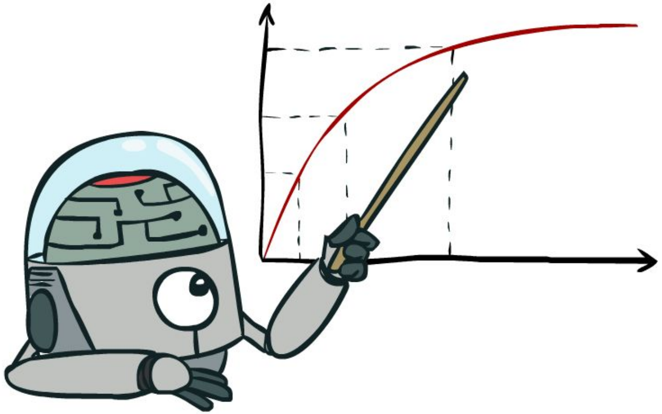
# Approach 3: Thinking Rationally

- This approach is about making machines **think logically**, based on reasoning and principles of logic rather than human imitation. It focuses on what is *correct* or *rational* rather than how humans actually think.
- **Example:**
- **Expert systems** that use logic rules (IF–THEN statements) to solve problems.
- **Mathematical theorem provers** that derive correct solutions through reasoning.
- **Chess programs** that calculate the best possible move logically.

# Approach 4: Acting Rationally

- This is the most **modern and practical approach**. It focuses on creating **rational agents** — systems that act to achieve the best outcome or expected result, given what they know. The agent may not always act like a human, but it acts *rationally* based on data and environment.
- **Example:**
- **Self-driving cars** making safe, optimal driving decisions.
- **Recommendation systems** (like Netflix or Amazon) suggesting products or movies intelligently.
- **AI game agents** that adapt their strategy to win.

# Maximize Your Expected Utility



# Foundations of AI

- Philosophy
  - 450 BC, Socrates asked for algorithm to distinguish pious from non-pious individuals
  - Aristotle developed laws for reasoning
- Mathematics
  - 1847, Boole introduced formal language for making logical inference
- Economics
  - 1776, Smith views economies as consisting of agents maximizing their own well being (payoff)
- Neuroscience
  - 1861, Study how brains process information
- Psychology
  - 1879, Cognitive psychology initiated
- Linguistics
  - 1957, Skinner studied behaviorist approach to language learning

# What About the Brain?

- Brains (human minds) are very good at making rational decisions, but not perfect
- Brains aren’t as modular as software, so hard to reverse engineer!
- “Brains are to intelligence as wings are to flight”
- Lessons learned from the brain: memory and simulation are key to decision making

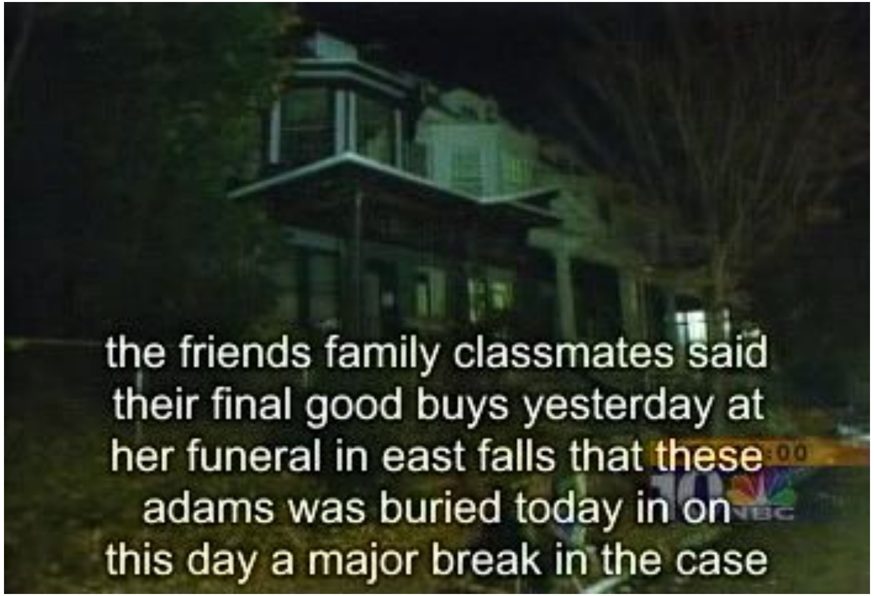
# What Can AI Do?

Quiz: Which of the following can be done at present?

- Play a decent game of table tennis?
- Play a decent game of Jeopardy?
- Drive safely along a curving mountain road?
- Buy a week's worth of groceries on the web?
- Discover and prove a new mathematical theorem?
- Converse successfully with another person for an hour?
- Perform a surgical operation?
- Put away the dishes and fold the laundry?
- Translate spoken Chinese into spoken English in real time?
- Write an intentionally funny story?

# Natural Language

- Speech technologies (e.g. Siri)
  - Automatic speech recognition (ASR)
  - Text-to-speech synthesis (TTS)
  - Dialog systems




Demo: NLP – ASR

# Natural Language

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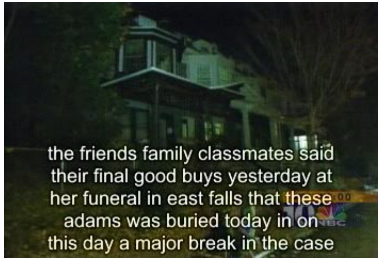




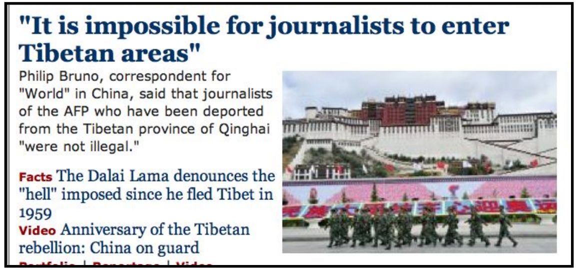
- Language processing technologies
  - Question answering
  - Machine translation

**"Il est impossible aux journalistes de rentrer dans les régions tibétaines"**

Bruno Philip, correspondant du "Monde" en Chine, estime que les journalistes de l'AFP qui ont été expulsés de la province tibétaine du Qinghai "n'étaient pas dans l'illégalité".

Les faits Le dalaï-lama dénonce l'"enfer" imposé au Tibet depuis sa fuite, en 1959

Vidéo Anniversaire de la rébellion



**"It is impossible for journalists to enter Tibetan areas"**

Philip Bruno, correspondent for "World" in China, said that journalists of the AFP who have been deported from the Tibetan province of Qinghai "were not illegal."

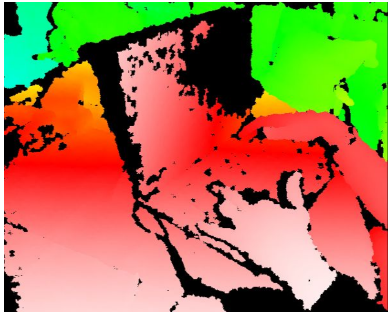
Facts The Dalai Lama denounces the "hell" imposed since he fled Tibet in 1959

Video Anniversary of the Tibetan rebellion: China on guard

  - Web search
  - Text classification, spam filtering, etc…

# Vision (Perception)

- Object and face recognition
- Scene segmentation
- Image classification

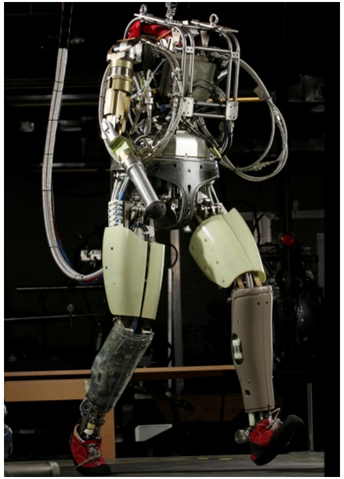



Images from Erik Sudderth (left), wikipedia (right)

# Robotics

- Robotics
  - Part mech. eng.
  - Part AI
  - Reality much harder than simulations!
- Technologies
  - Vehicles
  - Rescue
  - Soccer!
  - Lots of automation…
- In this class:
  - We ignore mechanical aspects
  - Methods for planning
  - Methods for control





Images from UC Berkeley, Boston Dynamics, RoboCup, Google

# Logic

- Logical systems
  - Theorem provers
  - NASA fault diagnosis
  - Question answering

- Methods:
  - Deduction systems
  - Constraint satisfaction
  - Satisfiability solvers (huge advances!)

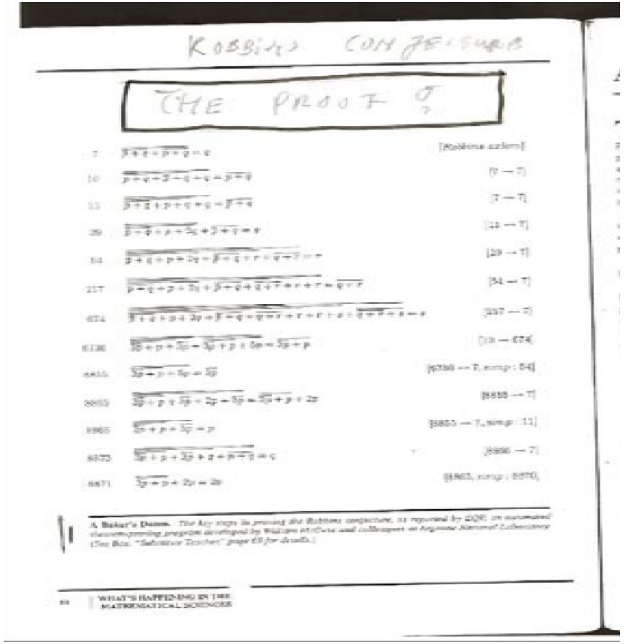
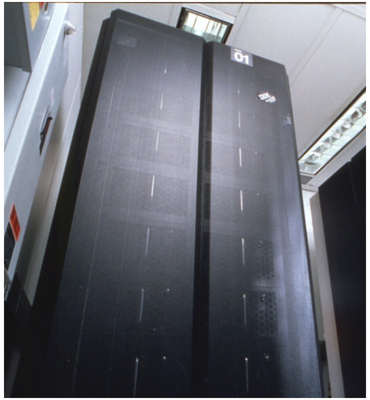



Image from Bart Selman

# Game Playing



- Classic Moment: May, '97: Deep Blue vs. Kasparov
  - First match won against world champion
  - “Intelligent creative” play
  - 200 million board positions per second
  - Humans understood 99.9 of Deep Blue's moves
  - Can do about the same now with a PC cluster
- Open question:
  - How does human cognition deal with the search space explosion of chess?
  - Or: how can humans compete with computers at all??
- 1996: Kasparov Beats Deep Blue
  “I could feel --- I could smell --- a new kind of intelligence across the table.”
- 1997: Deep Blue Beats Kasparov
  “Deep Blue hasn't proven anything.”
- Huge game-playing advances recently, e.g. in Go!

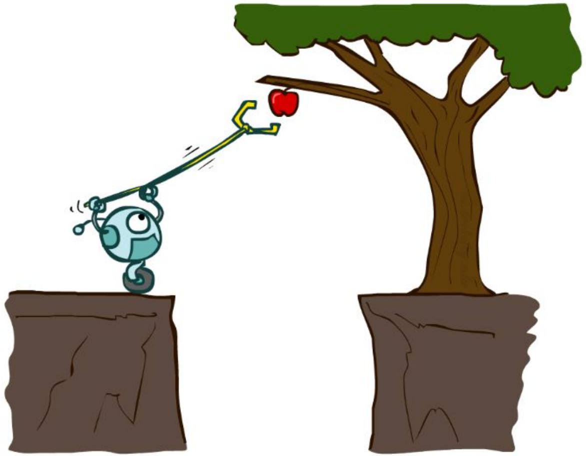

Text from Bart Selman, image from IBM’s Deep Blue pages

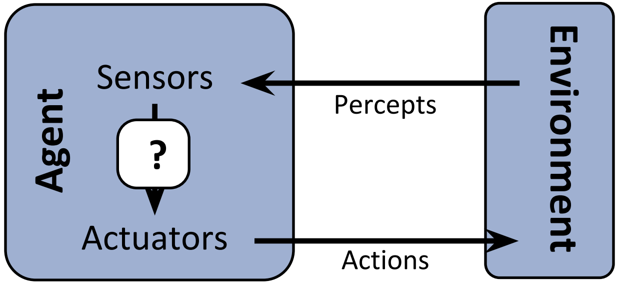
# Decision Making

- Applied AI involves many kinds of automation
  - Scheduling, e.g. airline routing, military
  - Route planning, e.g. Google maps
  - Medical diagnosis
  - Web search engines
  - Spam classifiers
  - Automated help desks
  - Fraud detection
  - Product recommendations
  - … Lots more!

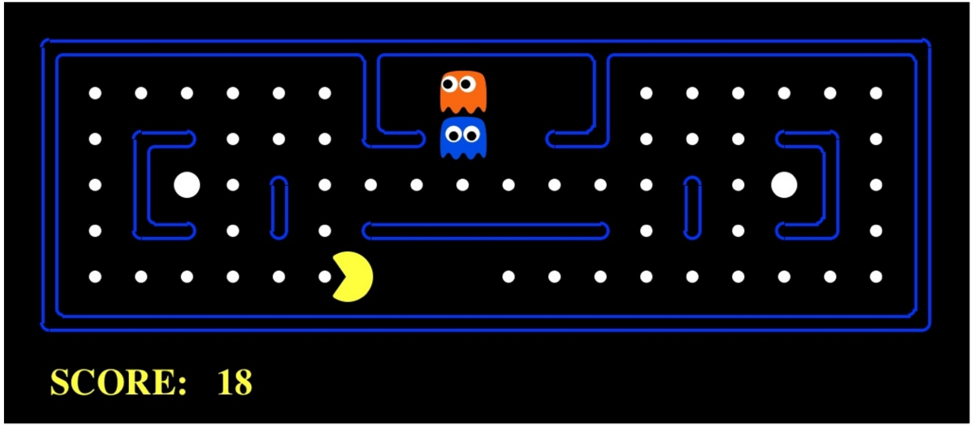
# Designing Rational Agents

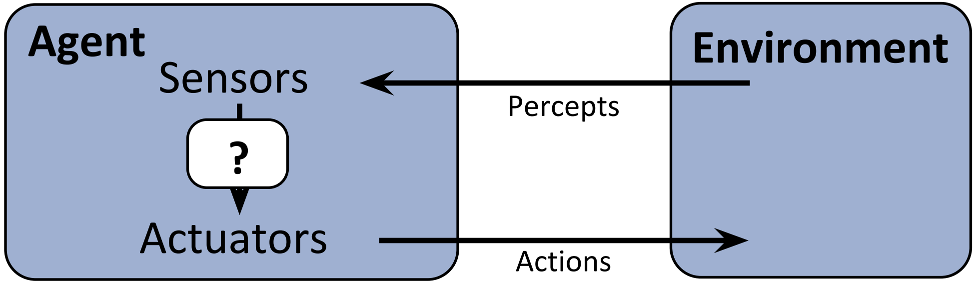
- An **agent** is an entity that *perceives* and *acts*.
- A **rational agent** selects actions that maximize its (expected) **utility**.
- Characteristics of the **percepts, environment,** and **action space** dictate techniques for selecting rational actions





# Pac-Man as an Agent




Pac-Man is a registered trademark of Namco-Bandai Games, used here for educational purposes